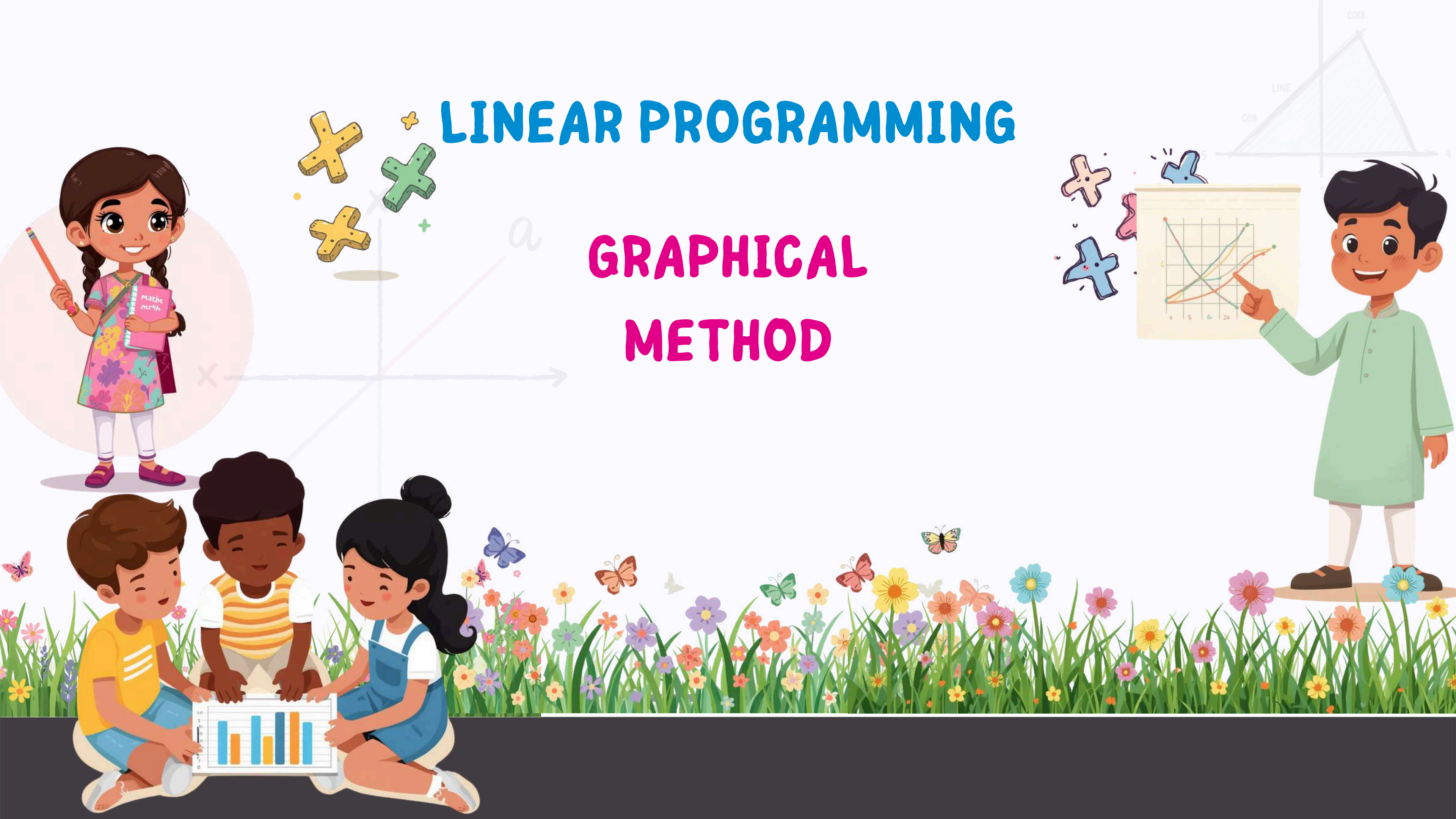


LINEAR PROGRAMMING

GRAPHICAL METHOD



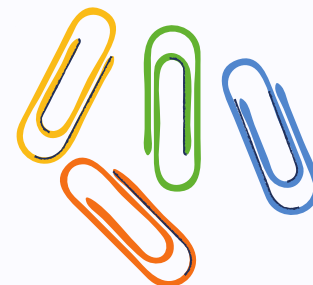


What is Linear Programming?

Linear programming helps us find the best solution using math!

It's a special way to solve problems where we need to make the smartest choice possible.

- ✔ Choose the best outcome (like maximum profit or minimum cost)
- ✔ Use math rules called constraints

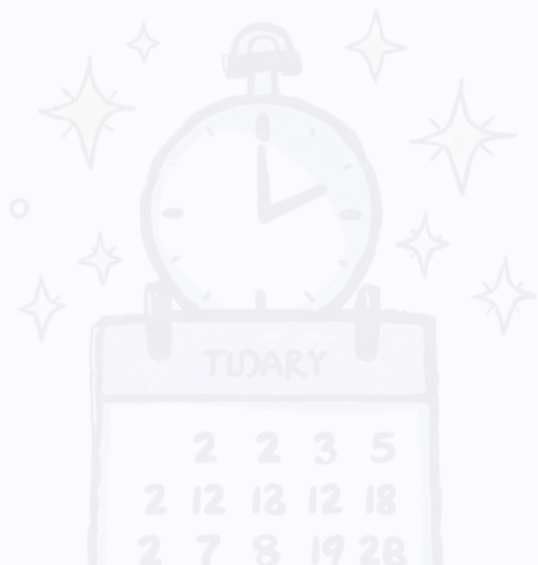
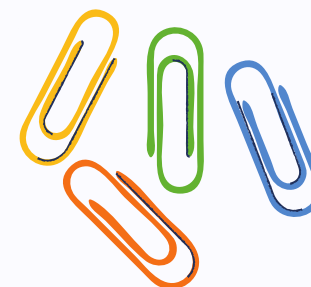




Why Do We Use Linear Programming?

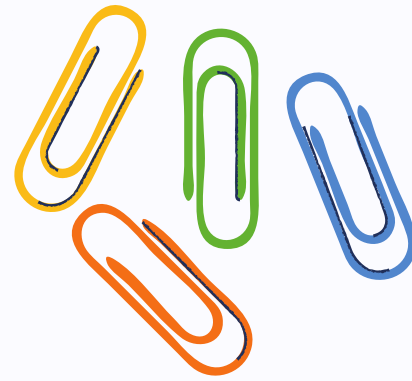
Linear programming helps us make smart choices every day! Whether we're planning a birthday party or deciding how to spend our pocket money, math helps us find the best answer.

Planning how to spend money or time wisely helps us get the most out of what we have!





Key Parts of Linear Programming



Objective Function

What we want to maximize or minimize, like getting the most profit or least cost.

Constraints

Rules we must follow, like limits on time, money, or resources.

Variables

What we can change or decide, like how many items to buy or make.

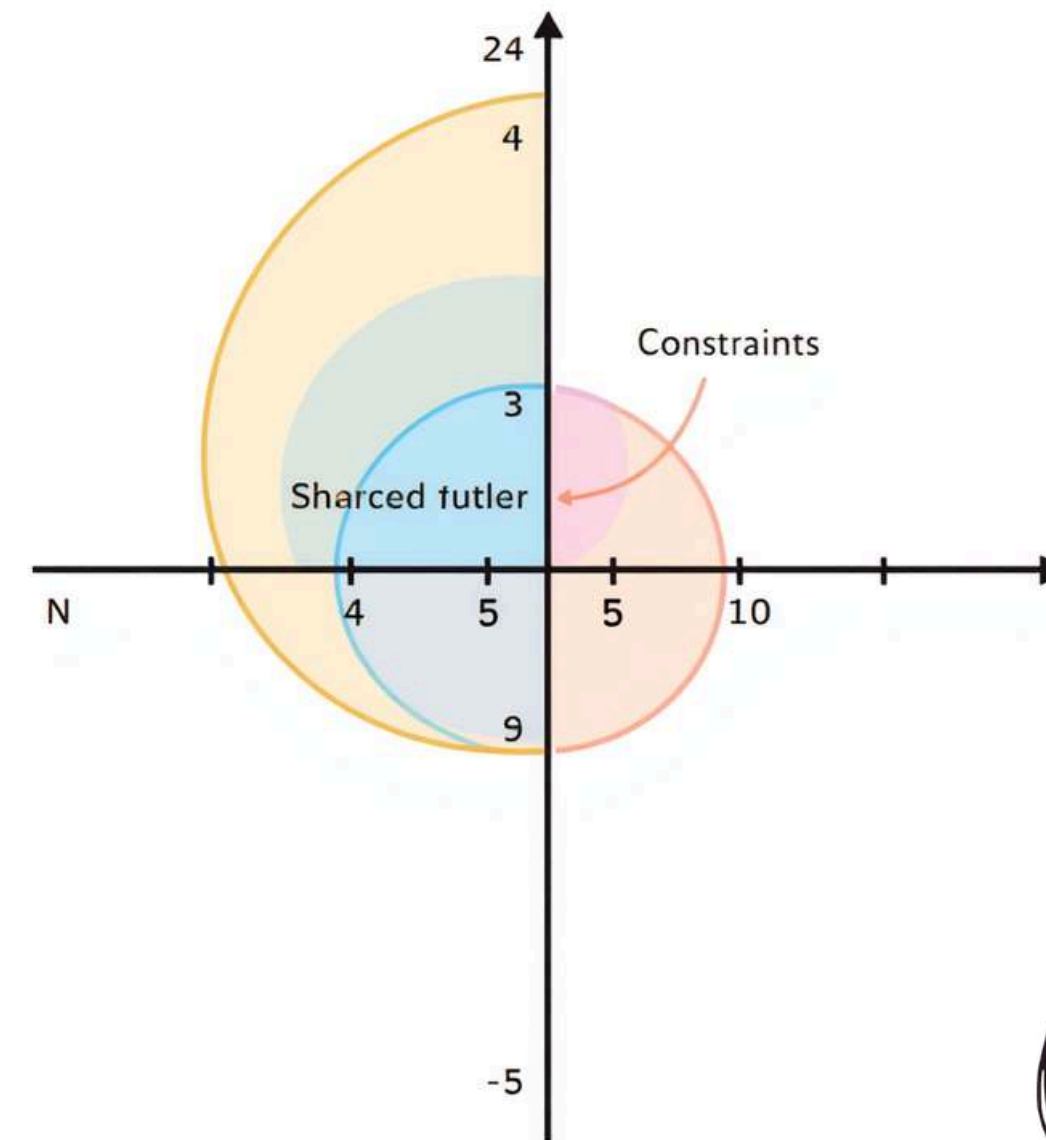




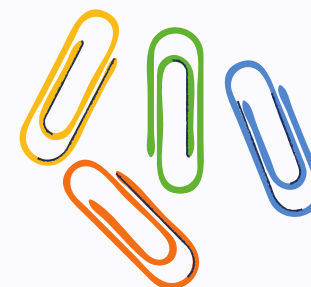
Understanding Constraints

Constraints are rules that tell us what is allowed! In linear programming, we write these rules using inequalities. They help us know the boundaries of our problem.

- ✓ $x + y \leq 10$ means the total must be 10 or less
- ✓ $x \geq 0$ and $y \geq 0$ means no negative values allowed



$$+cos + -y = 10$$





Graphing Inequalities

Step 1

Let's learn how to graph an inequality!

- First, draw the line $x + y = 10$ on the coordinate plane
- Use a solid line when you see $\leq$ or $\geq$ (includes the line)
- Use a dashed line when you see $<$ or $>$ (does not include the line)
- Plot two points to draw your line accurately





Graphing Inequalities

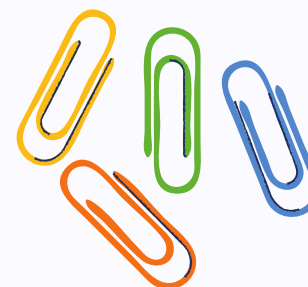
Step 2

Now we shade the correct side of the line!

How to find which side to shade:

- Pick a test point (like 0,0) not on the line
- Put the point into the inequality
- If it's true, shade that side
- If it's false, shade the other side

The shaded area shows all possible solutions!

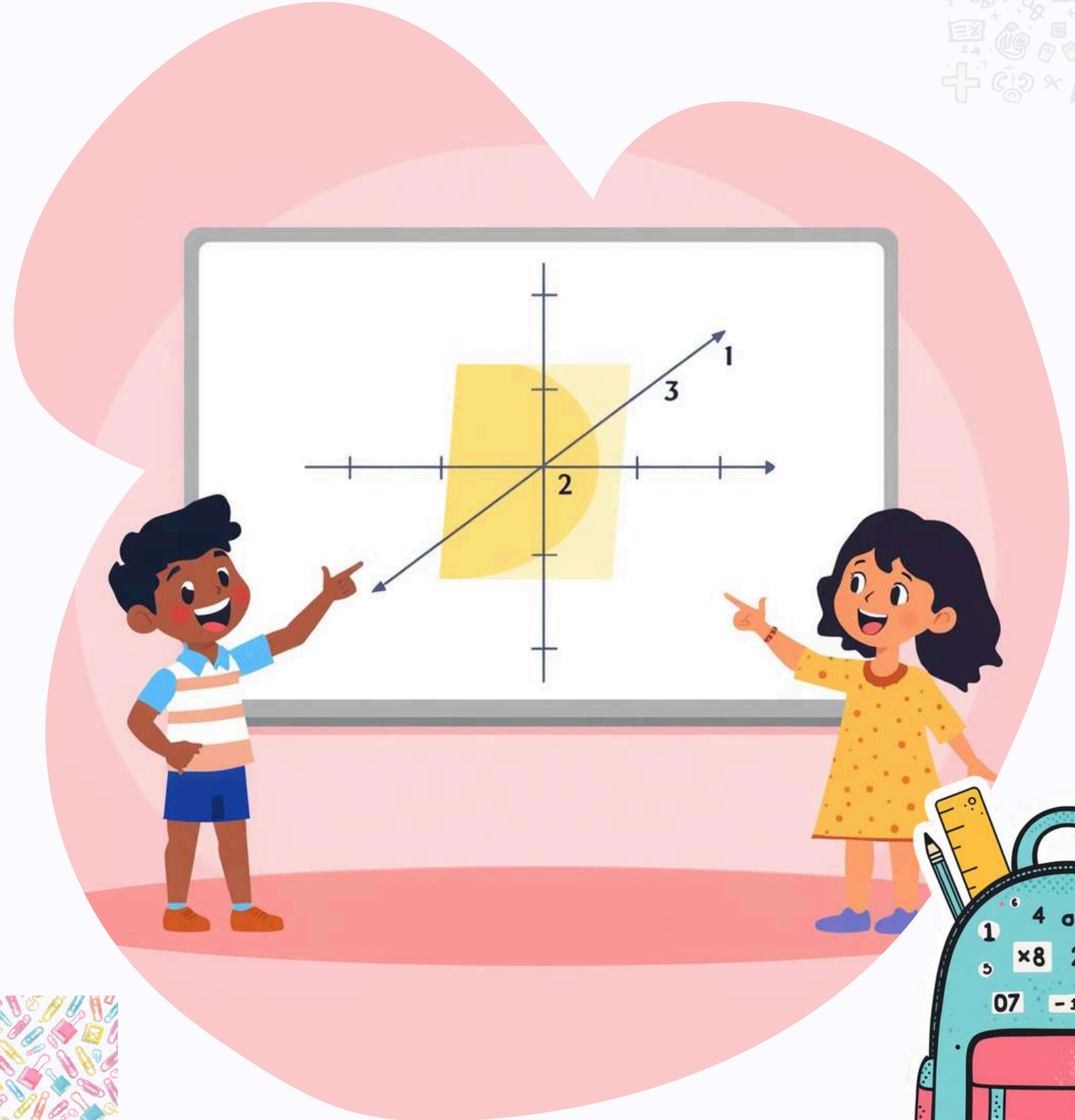




What is the Feasible Region?

The feasible region is the area on a graph where all constraints overlap. This special zone shows all the possible solutions that follow every rule!

Think of it like the sweet spot where everything works together. The best answer to our problem will always be found somewhere in this shaded area.

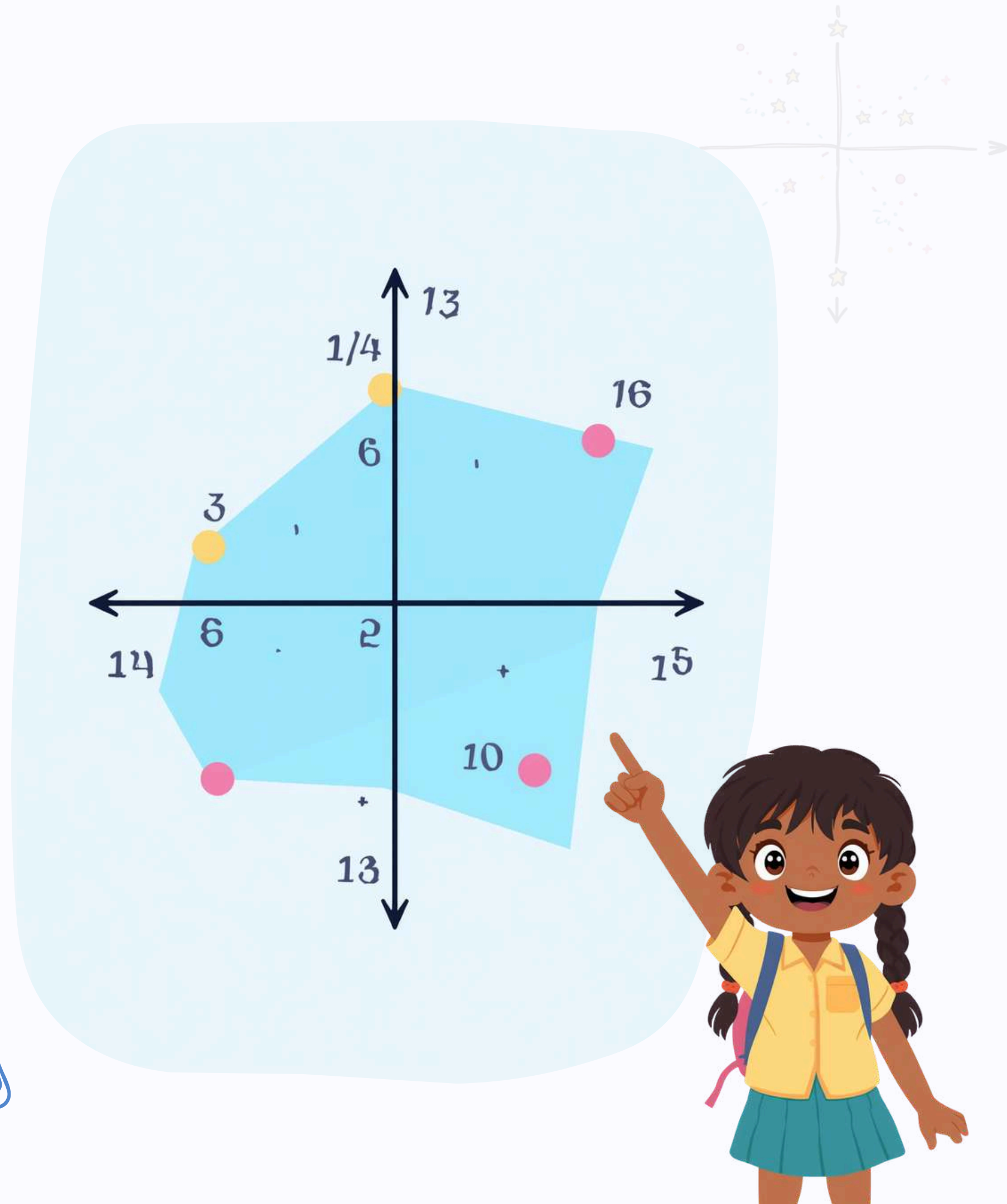
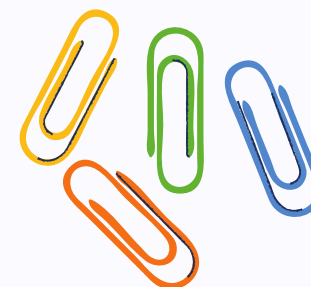
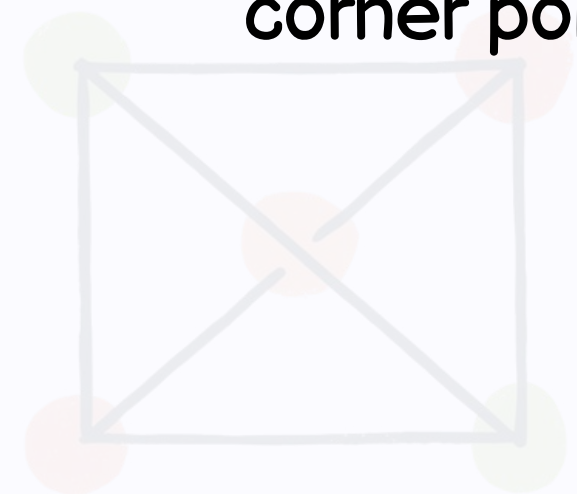




Finding Corner Points

Corner points are special spots where the boundary lines of our feasible region meet. These points are super important because the best solution to our linear programming problem is always found at one of these corners!

- ✔ Check all corners of the feasible region carefully.
- ✔ The best solution is always at one corner point!





Evaluating the Objective Function

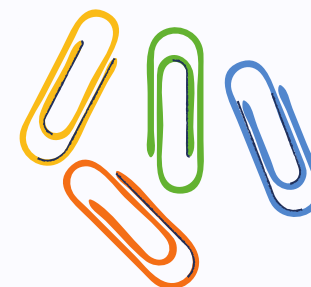
To find the best solution, calculate Z at each corner point!

Objective Function: $Z = 3x + 4y$

Corner Point Calculations:

- Point (0, 0): $Z = 3(0) + 4(0) = 0$
- Point (5, 0): $Z = 3(5) + 4(0) = 15$
- Point (0, 6): $Z = 3(0) + 4(6) = 24$
- Point (3, 4): $Z = 3(3) + 4(4) = 25$ ★ Best!

The maximum value is 25 at point (3, 4)!





Sample Problem

Problem Statement

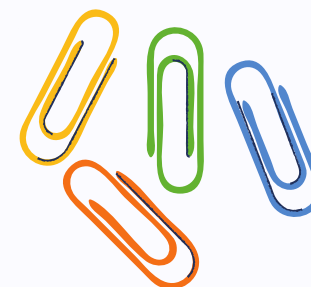
Let's Practice!

You have 10 candies and 12 cookies. You want to pack gift boxes with candies (x) and cookies (y). You want to maximize the total number of items in each box.

Constraints:

- $x + y \leq 10$ (candies limit)
- $x + y \leq 12$ (cookies limit)
- $x \geq 0$ and $y \geq 0$ (no negatives)

Objective: Maximize $Z = x + y$





Sample Problem

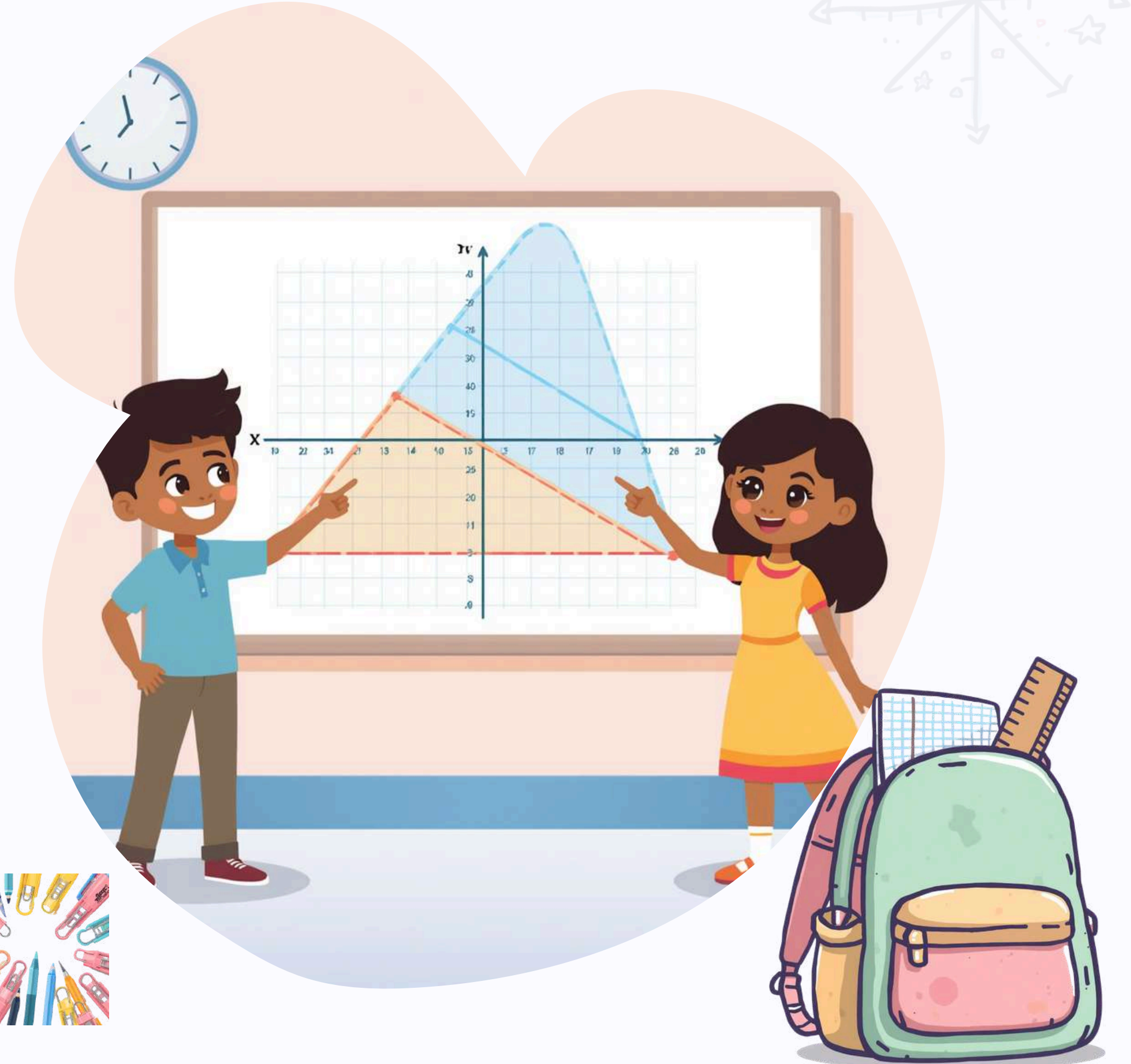
Graphing

Constraints

Let's graph our constraints on a coordinate plane:

- $x + y \leq 10$ (candies + cookies limit)
- $x \leq 10$ and $y \leq 12$ (available items)
- $x \geq 0$ and $y \geq 0$ (no negatives)

The shaded area where all constraints overlap is our feasible region — that's where our solution must be!





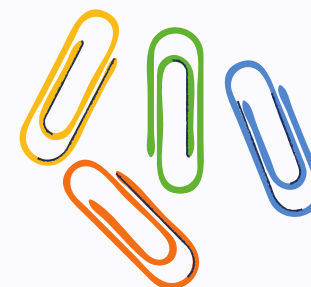
Finding the Best Solution

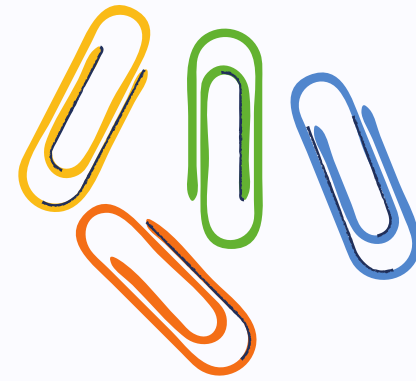
Now let's check each corner point!

Objective Function: $Z = x + y$

- Corner (0, 0): $Z = 0 + 0 = 0$
- Corner (10, 0): $Z = 10 + 0 = 10$
- Corner (0, 12): $Z = 0 + 12 = 12$
- Corner (10, 2): $Z = 10 + 2 = 12$

★ Maximum value is 12 at corners (0, 12) or (10, 2)!





Steps to Solve LP

Steps 1-2

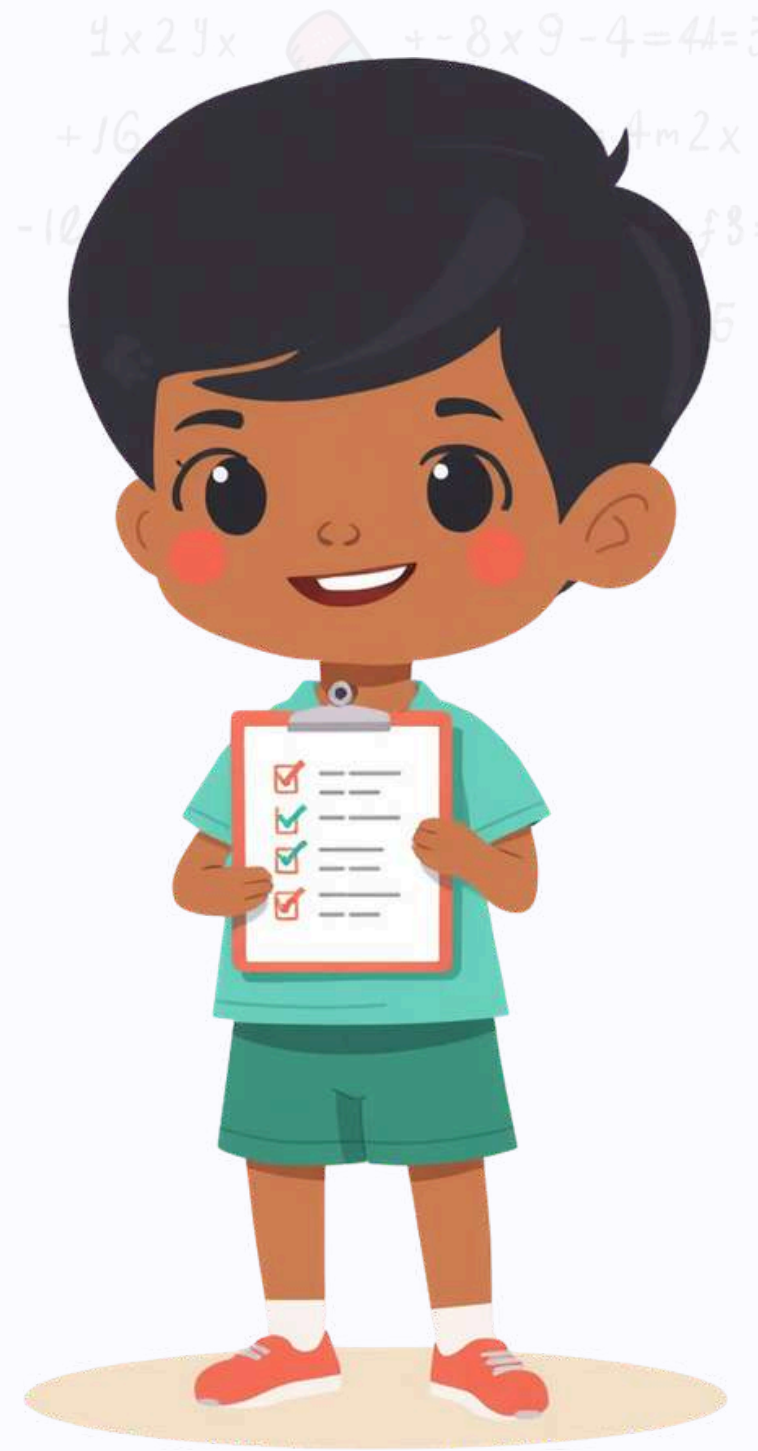
Write constraints and objective function.
Graph inequalities on coordinate plane.

Steps 3-4

Find the feasible region where all constraints overlap. Identify corner points.

Steps 5-6

Calculate objective function at each corner.
Choose the best value!

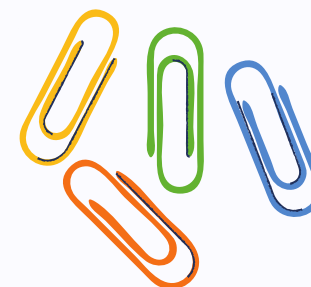




Where Can We Use Linear Programming?

Linear programming helps us make smart choices in everyday life! Here are some fun examples:

- Planning snacks for a party
- Scheduling time for homework and play
- Sharing toys fairly among friends



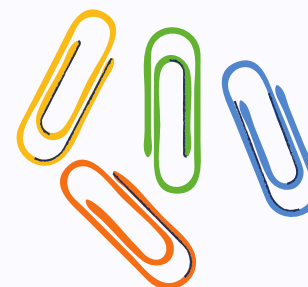


PractiTime!

Try this at home! Here's a fun problem to solve:
You have ₹20 to spend on pencils (x) and erasers (y). Pencils cost ₹3 each and erasers cost ₹2 each. You want to buy as many items as possible.

Objective: Maximize $Z = x + y$

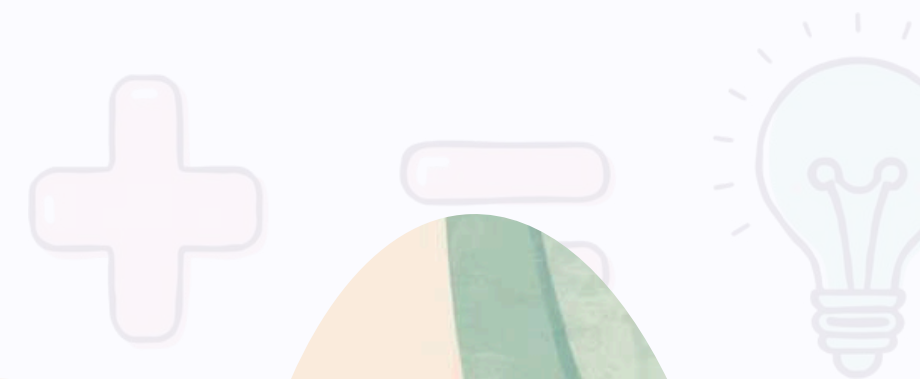
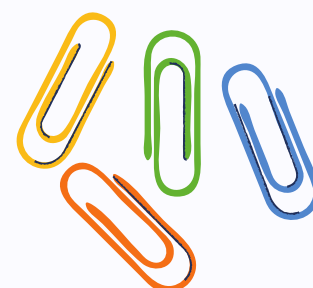
- ✔ Constraint 1: $3x + 2y \leq 20$ (budget limit)
- ✔ Constraint 2: $x \geq 0$ and $y \geq 0$ (no negatives)





Keep Exploring! Math Adventures Await You!

"Math helps us solve puzzles in real life!"
Every problem you solve makes you
stronger. Keep practicing, keep questioning,
and keep discovering the magic of
mathematics all around you!





THANK YOU!

Thank you for learning with us!



Presented by
Grade 6 Maths Team

